

Real Time Streaming of 8K 360 Degree Video to Mobile VR Headsets

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ABSTRACT

In this demo, we showcase how to stream 8K 360° video to commodity mobile devices without pre-processing or viewpoint prediction using our MEC-VR system. Users can freely select 360° video from YouTube and immediately become immersed in the video of 8K resolution using a Samsung GearVR compatible smartphone. Our system can support zoom in and zoom out while saving up to 80% bandwidth.

CCS CONCEPTS

• **Networks** → *Cloud computing; Mobile networks*; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

KEYWORDS

Mobile VR, Mobile Edge Cloud, 360 Degree Video

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1 INTRODUCTION

360° video is an emerging video technology that becomes more and more popular. Watching a 360° video, especially using a head-mounted VR display provides a more immersive experience compared to watching a normal video because the viewer is surrounded by the video and can proactively choose the preferred viewpoint. Streaming 360° video requires much more bandwidth than streaming the conventional video of the same quality. The pixels of 360° video spread in the sphere to capture views of all directions, but the video player only presents a portion of all pixels during playback when selecting a given viewpoint. According to [2], the 360 video of 8K resolution or above is necessary for a decent VR experience. Even though content providers like YouTube has already provided 360° video at 8K resolution, streaming and playing such video at full frame rate is beyond the capability of currently available mobile VR headsets, that rely solely on the mobile hardware to perform rendering and display frames.

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Researchers have proposed different methods to enable existing mobile hardware to play 360° video of 8K or higher resolution. For example, Rubicks [1] divides the full 8K video frame into smaller tiles and streams only the visible tiles to the user. Rondao et al. [3] achieved 16K 360° video streaming by downscaling the original video to different levels and creating a pyramid of video tiles at 4K resolution. The system allows the mobile user to zoom in for more details and zoom out for a wide angle view. Depending on the viewpoint and zooming scale, the mobile device only plays one video tile of 4K resolution at any time. However, both methods require the pre-processing of original 360° video to generate video tiles and depend on the history view trace data to predict user's future viewpoint. Incorrect viewpoint prediction usually leads to video stall, quality fluctuation and bad user experience.

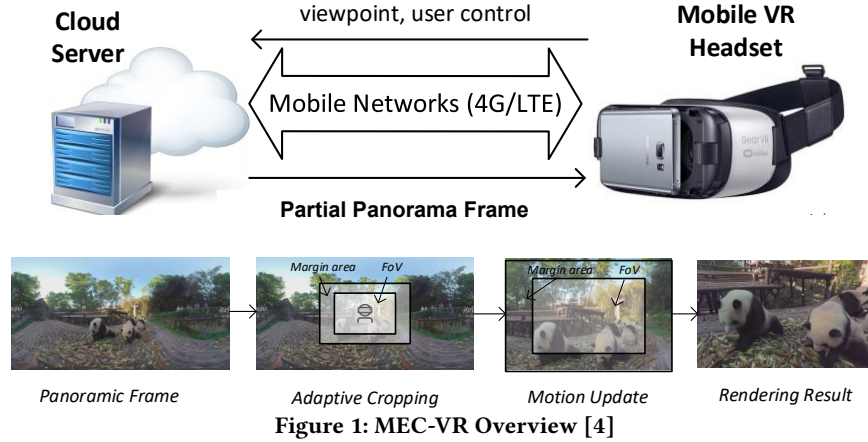
In this demonstration, we present a MEC-VR prototype system that streams any YouTube 8K 360° video to a mobile VR headset in real time with **no pre-processing** and **no viewpoint prediction**. We will invite the audience to try our demo application in person and experience the better quality provided by 8K 360° video. In addition, We will compare the visual quality and bandwidth usage of our system with the conventional streaming approach during the demonstration.

2 SYSTEM DESIGN

Our demonstration system is based on MEC-VR [4]. This work has been accepted to present in the main conference. Fig. 1 shows an illustration of how MEC-VR works. It utilizes the computing resources deployed in the Mobile Edge Cloud to help processing the 8K 360° video in real time. The server decodes the original video frame and only sends a sub-frame that is slightly larger than the currently user's view area to the mobile device. The mobile client uses the last received frame to perform local rendering for the latest viewpoint and at the same time notifies the server about the viewpoint change every time when the user moves the head. The low latency and adaptive margin size are the key factors to the success of MEC-VR. The client should receive a new frame before the viewpoint moves out of the area covered by the margin in the previous frame so that the user will not realize that the mobile device only gets a partial frame. Please refer to our full paper [4] for more details of MEC-VR.

We have also added some new components and features in this demo system.

YouTube Launcher: Our demo allows the user to select any YouTube 360° video for streaming. Therefore, we have created an HTTP service responsible for orchestrating the video processing for a given user request. The YouTube launcher takes the URL to



an YouTube video as input, uses youtube-dl¹ to retrieve the actual URL to the CDN server that hosts the request video.

Scaling and Zooming: We added a zooming feature to optimize streaming 8K content to small screen devices. For example, simply cropping the video frame at original 8K resolution may generate a sub-frame at 4K resolution or higher. However, the mobile VR headset can not display 4K video at full resolution on its small screen. Therefore, downscaling the 8K video first and generating a frame at 1440p or 1080p will have no influence on the final visual quality but save tremendously in computation and network bandwidth. Therefore, we add a scale parameter to control sub-frame generation. $scale = 1$ means no downscaling and $scale = 0.5$ means the server will downscale the video frame to half in width and height before cropping. This scaling feature also allows us to support zooming easily. We set the initial $scale$ value to 0.5 for 8K streaming and generate the sub-frame resolution of 1080p after cropping. If the user intends to zoom in and see more details of a video object, the system correlates the zoom-in gesture to the value of $scale$. By increasing $scale$ until it reaches 1, the server puts more pixels within the area of interest in the sub-frame without changing the frame resolution. As a result, we allow the user to view all the video details in the original 8K pixel quality but only uses the bandwidth to stream a 1080p video.

3 DEMONSTRATION

In our demonstration, we will invite the audience to use our VR application in person. The user may choose to either wear a Samsung GearVR headset which is driven by a Samsung Galaxy Note 8 smartphone, or use a Samsung Galaxy Tablet if he prefers not wearing a headset. The application comprises the following steps:

YouTube Video Selection: At the beginning of our demonstration, we encourage the user to go to YouTube app, freely select any 360° video of interests, and share the video URL to our MEC-VR app. Our app takes the YouTube URL as input and start retrieving video for real time processing.

Video Streaming: Once the video starts streaming, our app allows the user to switch between four different modes: (a) *full 1080p*: streaming the full 360° video at 1080p resolution directly from YouTube. The user expects to see no stall but low visual

quality; (b) *full 4K*: streaming the full 360° video at 4K resolution directly from YouTube. The user expects to see a better quality but the video may stall occasionally to re-buffer; (c) *no-scale 8K*: stream the full 360° video at 8K resolution to MEC-VR server and then crop the sub-frame (without scaling) for the mobile VR device. As for the MEC-VR server, we plan to deploy it on a laptop within the same WLAN with the mobile VR device. In order to simulate the deployment in edge cloud, we will use the traffic shaping software to increase the network latency between MEC-VR server and mobile VR device to around 80 ms, which matches the measurement in [4]. The user expects to see the video quality better than *full 4K* and experience no video stall during the playback; (d) *zooming 8K*: stream the full 360° video at 8K resolution to MEC-VR server and then apply $scale$ parameter before cropping. The user expects to see the video quality equivalent to *no-scale 8K* and no stall during playback. The user can also uses the remote controller or finger gesture to zoom in and zoom out. The user expects to see more details of smaller objects after zoom-in.

During the video streaming, we will encourage the user to look around even abruptly to test the responsiveness of the system. If the network condition allows, we will connect the mobile VR device to the MEC-VR server we deploy in a central cloud (Microsoft Azure) to compare how network latency plays a big role in user experience.

Performance Monitoring: We also plan to use an external display to present the actual bandwidth usage of the system. The user expects to see the bandwidth usage between mobile VR and MEC-VR server is significantly reduced compared to the bandwidth used to download original 8K video.

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¹<https://rg3.github.io/youtube-dl/>